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🦜🔗 LangChain

Explained under 1 minute!

Large Language Models (LLMs) like GPT-4 are powerful. But on their own, they're kind of like genius interns: smart, but forgetful, context-limited, and not great at doing real-world tasks like

- Calling an API
- Searching a DB
- Chaining logic across multiple steps

If you want to build something real like a chatbot that pulls in ***company docs***, ***books a calendar slot***, or helps with ***customer queries*** you'll quickly hit walls with just the raw model.

Enter ***Langchain***!

Langchain is a framework that helps you build actual apps using LLMs by letting them interact with **tools**, **memory**, and **data** in a structured way.

It's like giving your smart intern a notepad, a browser, access to files, and the ability to follow instructions properly. We need Langchain because LLMs alone can't:

- Remember previous chats (no memory)
- Look up external data (like your company's database or files)
- Break down complex tasks into smaller ones
- Handle multi-step workflows
- Chain together reasoning + action + validation

Langchain fills those gaps so you don't have to reinvent the wheel every time.

What it Does

Langchain helps you:

- Connect an LLM with memory (to remember chats, context)
- Plug into tools (APIs, search, calculators)
- Work with documents (PDFs, markdowns, Notion, etc.)
- **Build chains**: multi-step sequences that combine reasoning and action
- **Use agents**: LLMs that choose tools dynamically to solve a problem

How Does Langchain Work?

At its core, Langchain has these key building blocks:

- **Prompt Templates** – Predefined prompts with variables you fill in
- **Chains** – Sequences of calls to LLMs or tools (e.g., get a question → search docs → summarize → return)
- **Agents** – LLMs with tool access that decide what steps to take dynamically
- **Memory** – Keeps context between turns in a conversation
- **Tool integrations** – APIs, databases, search, web scraping, calculators, and more

It’s modular. So you can plug in your own LLM (OpenAI, Cohere, Anthropic), use different tools, and combine everything how you like.